Homework 2: Novel View Synthesis

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1 Scene and acquisition

The scene is a kitchen table with a moka pot, two steel cups, a glass, bananas and a plastic water bottle (glossy, metallic and transparent objects), next to a tiled white wall and a dark door. We took 41 phone photos (3072×4080) around the table at several heights and distances and kept the original files. Exposure time was constant (1/20 s) and ISO was 214 in 40 of the 41 photos (490 in one); white balance was automatic, so color shifts between photos are possible and add photometric error that the radiance field cannot explain geometrically. Camera poses come from a single COLMAP run (pycolmap, SIFT, exhaustive matching, one `SIMPLE_PINHOLE` camera) on all 41 images: all were registered (21,788 3D points). Five test images, spread over the capture and never used for training, were fixed once for all experiments; the other 36 form the baseline training set.

2 Training setup

All runs use nerfstudio's `splatfacto` (3D Gaussian Splatting), 30,000 steps, images downscaled by 2 (1536×2040), SH degree 3, seed 42 and otherwise default settings; only the list of training images changes between runs. The metric is PSNR, computed per image and then averaged over the set, with pixel values in $[0, 1]$ and on the downscaled images:

$$\mathrm{MSE} = \frac{1}{3N} \sum_{p=1}^{N} \sum_{c=1}^{3} \left(\hat{I}_c(p) - I_c(p) \right)^2, \qquad \mathrm{PSNR} = 10 \log_{10} \frac{1}{\mathrm{MSE}} \ \mathrm{dB}.$$

3 Baseline

Mean PSNR is 33.29 dB on the 36 training views and 24.34 dB on the 5 test views (Table 1), a gap of about 9 dB, i.e. roughly eight times the MSE. The gap is expected: the Gaussians are optimized to reproduce exactly the training photos, while a test view is rendered from a pose the field never fitted and also inherits imperfect geometry and view-dependent appearance. The best test view (090620_1, 27.6 dB) was shot in the same second as a training photo and is probably inflated; the other four average 23.5 dB. The two lowest views (090538: 22.4 dB, 090714: 22.8 dB) show the most reflective objects at close range, but Fig. 1 shows the largest errors on the background wall and door, so we cannot attribute them to specular surfaces alone. With only 5 test views and one run per setting, differences of a few tenths of a dB are not significant.

Run	Train img.	PSNR train	PSNR test	PSNR test (4 views*)
Baseline	36	33.29	24.34	23.53
3/4 of the views	27	34.90	23.68	22.84
1/2 of the views	18	36.37	20.85	21.14

Table 1: Mean PSNR (dB). *Excluding 090620_1, whose near-duplicate is in the training set of the first two runs only.

Test view	36 views	27 views	18 views
090538	22.39	21.55	18.28
090604	25.16	23.48	21.87
090620_1	27.56	27.03	19.69
090643	23.78	23.33	22.74
090714	22.79	23.01	21.68

Table 2: Test PSNR (dB) per view.

4 How many views are enough?

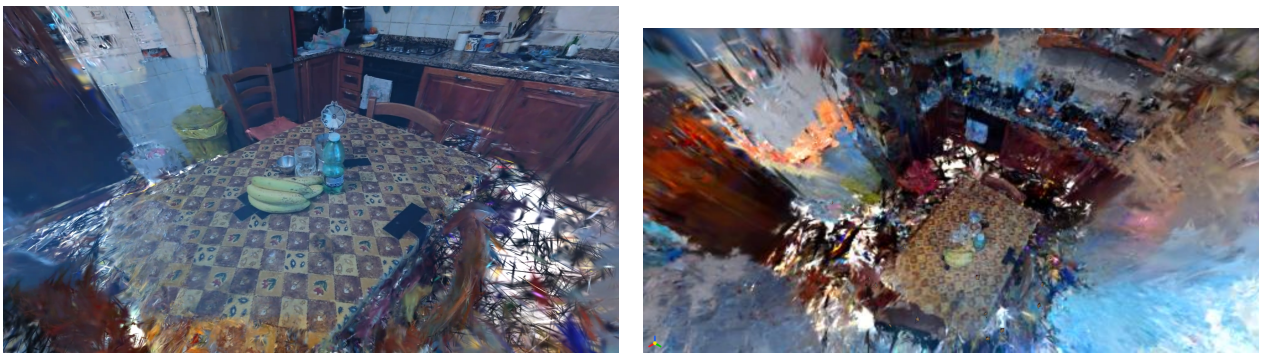
We retrained on subsets of 27 (3/4) and 18 (1/2) of the training images, with the same test set, steps and settings. The 27-view subset drops every fourth training image and the 18-view subset keeps every other one, so the sets are nested. Test PSNR is 20.85, 23.68 and 24.34 dB for 18, 27 and 36 views: the gain shrinks from +2.83 to +0.66 dB, while training PSNR rises as views are removed (36.37, 34.90, 33.29 dB), so the train–test gap grows from 9.0 to 15.5 dB. View 090620_1 drops to 19.7 dB with 18 views because its near-duplicate leaves the training set only in that run; excluding it, the means are 21.14, 22.84 and 23.53 dB, with the same trend (Tables 1–2). Figure 1 shows two test views for the three models: with fewer views the central objects remain recognizable, whereas errors concentrate on the wall, the door and the image borders, with streaks and floating Gaussians. These regions are seen from fewer directions, so the number of views needed is not uniform over the scene.



Figure 1: Two held-out views (rows): ground truth and renderings by the models trained on 36, 27 and 18 images, with PSNR. Test views, steps and settings are identical across runs.

5 A view between two cameras and outside coverage

Figure 2 (left) is the frame halfway between two training cameras (090627 and 090620, poses verified), rendered from the viewer at 1920×1080 with a 75° field of view. Table and objects are sharp, but streaks and dark needles appear near the borders, where few training rays likely constrain the Gaussians (the wide landscape frame also covers areas that the portrait training photos probably did not). Figure 2 (right) is rendered from far above the table, outside the captured region: the table remains recognizable, but the surrounding room dissolves into radial streaks, stretched Gaussians and color blobs. A splatting model has no generative prior: away from the training poses it shows whatever the optimization left unconstrained. Quality is therefore reliable only near the captured poses.



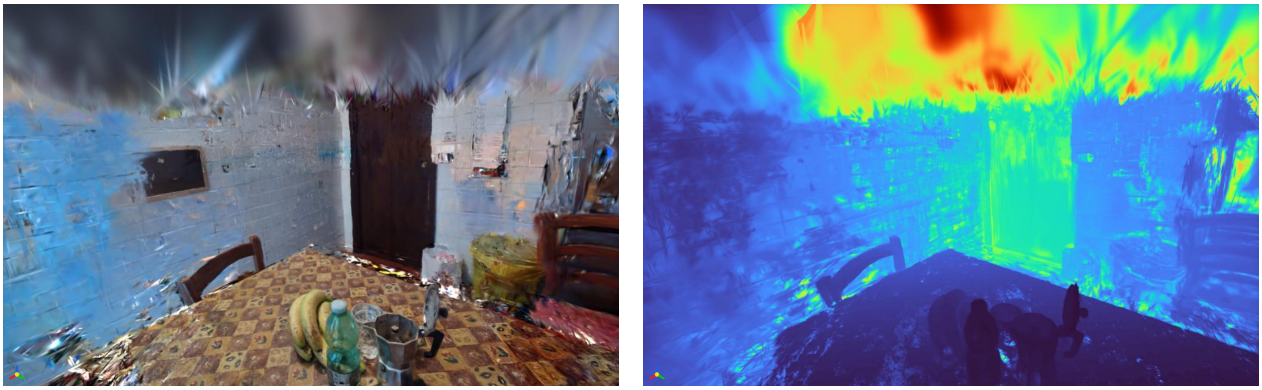
(a) Between two training cameras.

(b) From above the table, outside coverage.

Figure 2: Novel views of the baseline model.

6 Geometric distribution of the radiance field

Figure 3 shows a viewpoint looking at the corner of the room, as RGB and depth from the same pose (turbo colormap, normalized per frame, blue = near). Near the table the render is plausible, but the tiled wall is fragmented into blobs, spikes radiate from the top of the door, and in the depth map the door does not end on a surface: depth increases continuously upward. This viewpoint lies close to several training cameras, so the errors are not an extrapolation artifact. These regions are low-textured (uniform white tiles, a dark door) and most likely appear in the training photos only near the frame borders. The optimization minimizes photometric error on the training views only, and where texture is missing many depth assignments reproduce the same pixels; nothing penalizes Gaussians that stretch or drift along viewing rays until a new camera reveals it. The reconstruction is photometrically plausible from the training poses but geometrically wrong.



(a) RGB.

(b) Depth.

Figure 3: Geometric artifact above the door and on the tiled wall (baseline, same pose).

7 Conclusions

A radiance field behaves as an image interpolator constrained by view coverage: it is reliable near the captured poses, degrades away from them, and can look convincing where the geometry is wrong (low-texture or poorly covered surfaces). Removing views hurts non-uniformly, first in regions poor in texture or seen from few directions. With five test views and one run per setting, these are observations on a single scene, not general statistics.